Two years after the publication of his equation, Einstein decides to use it to describe the space of the entire universe, considered at the largest scale. And here he has another of his amazing ideas.

For thousands of years, men had asked themselves whether the universe was infinite, or had a limit. Both hypotheses entail thorny problems. An infinite universe does not seem to stand to reason: if it is infinite, for example, there must exist somewhere a reader just like you who is reading the very same book (infinity is truly vast, and there are not sufficient combinations of atoms to fill it with things always different from each other). In fact, there must be not only one but an infinite series of readers identical to yourself … But if there is a limit to the universe, what is that boundary? What sense is there in a border with nothing on the other side? Already in the fourth century CE, in Taranto, the Pythagorean philosopher Archytas had written:

> If I found myself in the furthest sky, that of the fixed stars, would I be able to stretch my hand, or a rod, out beyond it – or not? That I should not be able to is absurd; but if I am able to, then an outside exists, be it of matter, or space. In this way one could proceed ever further, towards the end, from time to time asking the same question, as to whether there will always be something into which to extend the rod.[4]

These two absurd alternatives – the absurdity of an infinite space, and the absurdity of a universe with a fixed border – didn't seem to leave any reasonable choice between them.

But Einstein finds a third way: the universe can be *finite* and at the same time *have no boundary*. How? Just as the surface of the Earth is not infinite but does not have a boundary either, where it 'ends'. This can happen, naturally enough, if something is curved: the surface of the Earth is curved. And in the theory of general relativity, of course, three-dimensional space can also be curved. Consequently, our universe can be finite but borderless.

On the surface of the Earth, if I were to keep walking in a straight line, I would not advance ad infinitum: I would eventually get back to the point I started from. Our universe could be made in the same way: if I leave in a spacecraft and journey always in the same direction, I fly around the universe and eventually end up back on Earth. A three-dimensional space of this kind, finite but without boundary, is called a 3-sphere.